The Thurstone Tilt Is a Schur Bridge, and Both of Its Ends Are Races

Peter Cotton*

September 20, 2026

Abstract

Thurstone portfolio polishing calibrates latent abilities that reproduce a benchmark under a reference dependence law and re-races them under a richer one, with a scalar ϕ dialling the estimated correlation in. Choosing the reference to be the block-diagonal part of the estimate makes that blend exactly the estimate with its cross-cluster entries scaled by ϕ, and for a two-block partition the conditional covariance of that matrix is the damped Schur complement at $\gamma = \phi^2$. The tilt dial is therefore the Schur dial under a square root. The pair of endpoints is new. Every bridge catalogued so far ends at $\Sigma^{-1}u$, a linear solve that can short any asset without bound; this one ends at a race, which lies on the simplex, splits a duplicated position rather than doubling it, and stays bounded. The taper is positive semidefinite for every ϕ, so unlike an unmatched pricing space the path has no pole, and the closed-form reliability of the cross-block estimate transfers as $\phi^\star = \sqrt{\gamma^\star}$, which sets the dial from data rather than by tuning.

1 Introduction

The bridges assembled on this site all end in the same place. Hierarchical risk parity, hierarchical equal risk contribution, nested clustered optimization and plain inverse variance are near ends, and turning a dial carries each of them to $\Sigma^{-1}u$: minimum variance for $u = \mathbf{1}$, maximum diversification for $u = \sigma$, the tangency direction for $u = \mu$ [1, 4, 5]. That far end is a linear solve. It can short an asset without bound, it doubles a position when an asset is duplicated, and it needs a matrix inverse.

Thurstone portfolio polishing [3] produces a portfolio of a different kind. Weights are read as the winning probabilities of a race among the assets, abilities are calibrated so that the race under a reference law reproduces a benchmark, and the same abilities are then re-raced under a richer law. The result is the gradient of an expected maximum, so it lies on the simplex by construction, it splits rather than doubles a duplicated position, and no covariance is inverted. In the Gaussian case the two laws are two correlation matrices and a scalar $\phi \in [0, 1]$ dials between them,

$$C_{\text{tilt}} = (1 - \phi)\, C_{\text{calib}} + \phi\, \widehat{C}, \tag{1}$$

with $\phi = 0$ reproducing the benchmark and $\phi = 1$ using the full estimate.

This note observes that (1) is a Schur damping for the right reference, so the tilt is a bridge in the sense of the other papers, and that its far end is a race rather than a linear solve.

*Microprediction. Email: peter.cotton@microprediction.com. A companion verification script is `paper/verify_tilt_bridge.py` at `github.com/microprediction/schur`. The identities below are checked there in exact rational arithmetic.

2 The reference that makes the blend a taper

Fix a partition of the assets into clusters and write $\ell(i)$ for the cluster of asset i. The *block taper* of a correlation $\widehat{C}$ at strength ϕ leaves the diagonal blocks alone and scales everything else:

$$\left(\widehat{C} \circ T_\phi\right)_{ij} = \begin{cases} \widehat{C}_{ij}, & \ell(i) = \ell(j), \\ \phi\, \widehat{C}_{ij}, & \text{otherwise.} \end{cases} \tag{2}$$

Proposition 1 (The blend is a taper)**.** *Let C_{calib} be the block-diagonal part of $\widehat{C}$, that is $\widehat{C}_{ij}$ when $\ell(i) = \ell(j)$ and zero otherwise. Then for every $\phi \in [0, 1]$ the blend* (1) *equals the block taper* (2)*.*

Proof. Within a diagonal block both matrices carry $\widehat{C}_{ij}$, since $(1 - \phi)\widehat{C}_{ij} + \phi\widehat{C}_{ij} = \widehat{C}_{ij}$. Off the diagonal blocks C_{calib} vanishes and the blend leaves $\phi\widehat{C}_{ij}$. □

The choice of reference matters here. The two references named in [3] are the identity and a one-factor market correlation. Under the identity the blend damps the within-cluster correlations as well, and is not (2); the certificate exhibits the difference. The block-diagonal reference is a third choice, and it is the one that aims the damping at the cross-cluster block alone.

3 The taper is the dial

Proposition 2 (Two blocks)**.** *Let the partition have two clusters A and B and let $\widehat{C} \succ 0$. The conditional covariance of A given B under the tapered correlation is the damped Schur complement of [1] at $\gamma = \phi^2$:*

$$\left(\widehat{C} \circ T_\phi\right)_{AA} - \left(\widehat{C} \circ T_\phi\right)_{AB} \left(\widehat{C} \circ T_\phi\right)_{BB}^{-1} \left(\widehat{C} \circ T_\phi\right)_{BA} \;=\; \widehat{C}_{AA} - \phi^2\, \widehat{C}_{AB} \widehat{C}_{BB}^{-1} \widehat{C}_{BA}. \tag{3}$$

Proof. With two clusters the taper leaves $\widehat{C}_{AA}$ and $\widehat{C}_{BB}$ unchanged and multiplies $\widehat{C}_{AB}$ by ϕ. Substituting gives $\widehat{C}_{AA} - \phi\widehat{C}_{AB}\widehat{C}_{BB}^{-1}\phi\widehat{C}_{BA}$. □

This is the identity of the localization note [6] read in the other direction. There a block-constant taper of an ensemble covariance was shown to have a damped complement as its conditional; here the same fact says that the tilt's blend and the Schur dial are one dial, related by $\gamma = \phi^2$.

Remark 1 (Beyond two blocks)**.** *With three or more clusters the taper also damps the cross-block entries* inside *the complement B, so $\left(\widehat{C} \circ T_\phi\right)_{BB}$ is no longer $\widehat{C}_{BB}$ and* (3) *becomes an approximation. On random correlations of twelve assets in three clusters the two sides differ by about 2×10^{-5} at $\phi = \frac{1}{2}$, against exact equality for two clusters. The flat construction of [4], which conditions each cluster on the others without damping inside them, is the exact multi-cluster analogue.*

4 Both ends are races

At $\phi = 0$ every cross-cluster correlation is zero, the clusters are independent, and the race decouples into one race per cluster with the benchmark reproduced inside each. At $\phi = 1$ the taper is the estimate itself and the race is run under the full correlation. Both ends are values of $W_S(\theta) = \nabla G_S(\theta)$ with $G_S(\theta) = \mathbb{E} \max_i(\theta_i + \eta_i)$ [3], so both are winning-probability vectors.

That is the substance of the title. A bridge whose far end is $\Sigma^{-1}u$ inherits the far end's pathologies at every γ near one: unbounded shorts, a doubled position under duplication, and an inverse that must exist. A bridge whose far end is a race inherits the opposite. The whole path

lies in the simplex, a duplicated asset splits its weight at every ϕ, and nothing is inverted. The dial still governs how much of the cross-cluster estimate is believed; what changes is what belief is fed into.

Corollary 1 (No pole)**.** *T_ϕ is positive semidefinite for every $\phi \in [0,1]$, so $\widehat{C} \circ T_\phi$ is positive semidefinite by the Schur product theorem and the race is well posed along the whole dial.*

Proof. $T_\phi = \phi\, J + (1-\phi)\,\mathrm{blockdiag}(J)$ with J the all-ones matrix, a nonnegative combination of two positive semidefinite matrices. □

The contrast is worth stating, because the other direction of generalization does not enjoy it. Pricing a cluster against something other than the portfolio held in it makes the outer $k \times k$ matrix asymmetric, and it can pass through singularity inside the dial [7]. Damping the law the race is run under cannot: the taper is a Schur product, and positive semidefiniteness is closed under it.

5 Setting the dial

Because $\gamma = \phi^2$, the closed-form reliability of the cross-block estimate carries over. Writing ρ for the conditional correlation across the block and n for the number of observations, [2] gives

$$\gamma^\star = \frac{(n-2)\rho^2}{(n-2)\rho^2 + (1-\rho^2)}, \qquad \text{hence} \qquad \phi^\star = \sqrt{\gamma^\star}. \tag{4}$$

The tilt strength is then read off the data rather than tuned, with the sample correlation r plugged in for ρ. The plug-in is a heuristic rather than an optimal shrinkage: under a true correlation of zero, $(n-2)r^2$ converges to a χ^2_1 variable rather than to zero, so $\gamma^\star$ keeps about a third of pure-noise coupling however large n, at a variance cost of order $1/n$ since the null cross-block estimate is itself of order $1/\sqrt{n}$. The square root matters: a coupling one would damp to $\gamma^\star = \frac{1}{4}$ on the covariance side corresponds to $\phi^\star = \frac{1}{2}$ on the tilt side, so a tilt that looks mild is a heavier damping than it appears.

What the evidence says. On block-structured markets with the covariance estimated from returns, the block taper and the global tilt of the identity reference behave differently: at two hundred and fifty observations the block taper is the better dial, and at sixty observations neither beats racing the clusters independently. This is the same pattern the covariance-side experiments show, and the reason is the one (4) encodes. The cross-cluster correlations are the part of the estimate worth least, so a dial aimed at them is the dial worth having.

6 Closing

Thurstone polishing was presented as a tilt by a view on dependence, and the Schur papers were presented as a bridge between a heuristic and an optimizer. With the block-diagonal reference they are one construction. The tilt's ϕ is the Schur γ under a square root, its near end is the independent cluster race, and its far end is the full race.

The far end is what makes this worth recording. Every other bridge here reaches a linear solve, and the interest lies in stopping short of it. This one reaches a portfolio that was already wanted for its own sake, so both ends are places one might stand, and the dial chooses between two defensible answers rather than between a heuristic and an optimum.

Two questions remain. The multi-cluster identity is exact only under the flat conditioning of [4], and whether the cheaper taper's error matters in practice is untested. And (4) is derived for a quadratic loss; whether it is the right strength when the consumer is a race, whose loss is the conjugate of an expected maximum, is open.

References

[1] P. Cotton. Schur complementary allocation: a unification of hierarchical risk parity and minimum variance portfolios. arXiv:2411.05807, 2024.

[2] P. Cotton. Two sides of Schur damping: high-dimensional pseudo-likelihoods and portfolio allocation. arXiv:2606.14798, 2026.

[3] P. Cotton. Thurstone portfolio polishing: tail-sensitive Black–Litterman and beyond. Working paper, 2026.

[4] P. Cotton. Nested clustered optimization is one end of a Schur bridge, and the interior is sometimes provably better. SSRN 7480738, 2026.

[5] P. Cotton. Hierarchical equal risk contribution is a corner of a Schur square, and two dials lead to minimum variance. Working paper, 2026.

[6] P. Cotton. Covariance localization is Schur damping, and its optimal strength is a sampling-error correction. Working paper, 2026.

[7] P. Cotton. The far end of a Schur bridge does not care how its clusters are priced, and the interior does. Working paper, 2026.